

SUNDARAM FINANCE LIMITED

Internal Tools & Technology Division

Form 26AS TDS Reconciliation System

Internship Project Report

Prepared by	Sanjana Ravishankar
Role	Full-Stack Developer Intern
Department	Data Platform
Organization	Sundaram Finance Limited
Technology Stack	Python · Flask · MySQL · HTML/CSS/JS · Power BI
Project Period	May – July 2026
Report Date	July 2026

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1. Project Overview

The Form 26AS TDS Reconciliation System is an internal web-based application developed during the summer internship at Sundaram Finance Limited. The system automates the reconciliation of Tax Deducted at Source (TDS) credits reported in TRACES Form 26AS against the company's Books of Accounts — a process that was previously performed manually using spreadsheets.

The application provides a comprehensive seven-screen interface covering Customer Master management, Form 26AS data viewing, Books of Accounts analysis, file upload management, automated reconciliation, TDS summary reporting, and a complete audit trail. In addition, a Power BI dashboard connected to the underlying MySQL database provides executive-level visual analytics.

The system was built entirely from scratch over the course of the internship, iteratively developed and tested with real-world data scenarios. All data processing, financial year detection, duplicate prevention, and reconciliation logic was implemented in Python using the Flask web framework, with a MySQL database backend and a responsive HTML/CSS/JavaScript frontend.

2. Problem Statement

Tax Deducted at Source (TDS) reconciliation is a critical compliance activity for large financial organizations. Sundaram Finance Limited receives TDS credits from multiple deductors across financial years, which must be matched against the company's internal income records to ensure accurate tax filings and identify discrepancies.

The manual reconciliation process suffered from several issues:

- Time-consuming manual comparison between Form 26AS PDF downloads and Excel-based Books of Accounts
- High risk of data entry errors and version mismatches between files
- No systematic tracking of which records had been reconciled, which were pending, and which had variances
- No centralized audit trail of who uploaded what and when
- Difficulty in identifying customers with under-credited or over-credited TDS amounts
- No historical financial year comparison or drill-down capability

This project was initiated to replace the manual process with an automated, auditable, and scalable internal tool that could handle multiple customers, multiple financial years, and multiple file formats in a single unified interface.

3. Objectives

- Build a full-stack web application for TDS reconciliation using Python Flask, MySQL, and modern HTML/CSS/JS
- Automate the parsing of Form 26AS PDFs (TRACES format) and XLSX/TXT Books of Accounts files
- Implement an intelligent reconciliation engine that auto-classifies records as Reconciled, Likely Match, Suggested Match, or Open based on match score percentage
- Provide a Customer Master module to centrally manage PAN, TAN, GSTIN, and customer status
- Build a complete Upload Utility with file hash-based deduplication, per-row financial year detection, and a full upload history log
- Design a TDS Summary report that consolidates TDS deductible versus Form 26AS credit per customer and section
- Implement an Audit Trail module for complete traceability of all upload and reconciliation events
- Connect the MySQL database to Power BI to deliver an executive-level visual dashboard with drill-through capability

4. Technology Stack

Backend	Python 3.x, Flask (micro web framework)
Database	MySQL 8.0 with 10+ tables, 4 views, and 11 performance indexes
Cloud Analytics (warehouse)	Google Cloud BigQuery (dataset: tds_reconciliation, project: eternal-skyline-468312-g7) GCP Auth Service Account JSON key with BigQuery Data Editor role
Frontend	HTML5, CSS3, Vanilla JavaScript (no frontend framework)
PDF Parsing	pypdf (text-layer extraction with TAN space-split correction)
Excel Parsing	openpyxl (read_only streaming mode for memory efficiency)
Reporting	Power BI Desktop connected via MySQL ODBC Connector
Development	VS Code, MySQL Workbench
Version	v1.0 — Internal

5. System Architecture

The application follows a classic three-tier architecture. The presentation tier consists of seven HTML pages served by Flask's Jinja2 templating engine, styled with a consistent navy-and-white corporate theme. The application tier is a Flask backend (app.py) containing all business logic — file parsing, FY detection, reconciliation engine, and API endpoints. The data tier is a MySQL database comprising ten operational tables, four Power BI views, and an optimized index set.

The data flow for a typical reconciliation cycle is:

- Customer profiles (PAN, TAN, GSTIN) are loaded via Customer Master upload
- Books of Accounts are uploaded as XLSX or TXT; each row is parsed, FY auto-detected from document date, and stored in `income_records`
- Form 26AS PDFs are uploaded and parsed; TAN extracted with space-split correction; transaction rows stored in `form26as_records`
- The reconciliation engine joins `income_records` and `form26as_records` by customer TAN, computes System TDS vs 26AS Credit, calculates variance and match score, and classifies each record
- Results are displayed in the Reconciliation screen; users can manually override status via the Change action
- All events are logged in `audit_logs`; all uploads recorded in `file_uploads_log`
- Power BI connects to four MySQL views for live dashboard refresh

6. Screen-by-Screen Implementation

6.1 Customer Master

The Customer Master is the foundational master data module. It stores and manages all customer profiles including Customer Code, Customer Name, PAN Number, GSTIN, TAN Number, and Active/Inactive status. The TAN number is the primary identifier used by the reconciliation engine to match Books of Accounts records against Form 26AS credits.

The screen displays three KPI cards at the top — Total Customers, Active Customers (with percentage of total), and Total TANs (with unlinked TAN count). The Customer Data tab shows a paginated, searchable list of all customers. Clicking any row opens a side panel with full customer details.

6.2 Form 26AS (TDS & TCS Sections)

The Form 26AS screen provides a read-only view of all parsed TDS and TCS records from uploaded TRACES PDFs. It is divided into two tabs — TDS Records and TCS Records — each showing the complete parsed data including GL Code, Section, Sub Code, Rate (%), Surcharge (%), Cess (%), Application Form, Applicable To, Description, and Effective From date.

Records can be filtered by entities and section. The screen also shows aggregate counts — TDS Records: 69 across all sections, TCS Records: 13 across all sections.

Form 26AS (TDS and TCS Sections)

View TDS and TCS records. Filter by payee type.

Filters: Applicable To: All [Apply] [Reset]

TDS RECORDS: 69 across all sections

TCS RECORDS: 13 across all sections

Search by GL code, section, description... [All Sections]

GL CODE	SECTION	SUB CODE	RATE (%)	SURCHARGE (%)	CESS (%)	APPLICATION FORM	APPLICABLE TO	DESCRIPTION	EFFECTIVE FROM
1	192	-	0.00%	0.00%	0.00%	Form 16	ALL	Payment of salary	2024-04-01
2	192A	-	10.00%	0.00%	0.00%	Form 16A	ALL	Payment of accumulated balance of provident fund taxable in hands of employee	2024-04-01
3	193	a	10.00%	0.00%	0.00%	Form 16A	ALL	Interest on debentures/securities by local authority or corporation	2024-04-01
4	193	b	10.00%	0.00%	0.00%	Form 16A	ALL	Interest on listed debentures issued by a company	2024-04-01
5	193	c	10.00%	0.00%	0.00%	Form 16A	ALL	Interest on Central/State Government security (8% Savings Bonds etc.)	2024-04-01
6	193	d	10.00%	0.00%	0.00%	Form 16A	ALL	Interest on any other security	2024-04-01
7	194	-	10.00%	0.00%	0.00%	Form 16A	ALL	Income by way of dividend	2024-04-01
8	194A	-	10.00%	0.00%	0.00%	Form 16A	ALL	Income by way of interest other than interest on securities	2024-04-01
9	194B	-	30.00%	0.00%	0.00%	Form 16A	ALL	Income by way of winnings from lotteries, crossword puzzles, card games, gambling	2024-04-01
10	194BA	-	30.00%	0.00%	0.00%	Form 16A	ALL	Income by way of winnings from any online game	2024-04-01

Showing 1-10 of 69 records

Figure 3: Form 26AS — TDS Records tab showing section codes with applicable rates

Form 26AS (TDS and TCS section codes)

Form 26AS (TDS and TCS Sections)
View TDS and TCS records. Filter by payee type.

Filters: Applicable To: All [Apply] [Reset]

TDS RECORDS
69 across all sections

TCS RECORDS
13 across all sections

Search by QL code, section, description... [All Sections]

QL CODE	SECTION	SUB CODE	RATE (%)	SURCHARGE (%)	CESS (%)	APPLICATION FORM	APPLICABLE TO	DESCRIPTION	EFFECTIVE FROM
GL-5201	394	--	2.00%	0.00%	0.00%	Form 27D	Scrap Buyer Profile	Collection on manufacturing scrap clearance	Wed, 01 Apr 2020 00:00:00 GMT
GL-5205	ABOLISHED	--	0.00%	0.00%	0.00%	Form 27D	Bulk Goods Purchase	Exempted - Former 206C(94) removed by 2025 Act	Wed, 01 Apr 2020 00:00:00 GMT
GL-5201	394	--	2.00%	0.00%	0.00%	Form 27D	Timber Merchant	Timber contract sale clearance allocation	Wed, 01 Apr 2020 00:00:00 GMT
GL-5205	ABOLISHED	--	0.00%	0.00%	0.00%	Form 27D	Hardware Logistics	Exempted - Former 206C(94) removed by 2025 Act	Wed, 01 Apr 2020 00:00:00 GMT
GL-5201	394	--	2.00%	0.00%	0.00%	Form 27D	Mineral Contractor	Collection on mining ore material clearance	Wed, 01 Apr 2020 00:00:00 GMT
GL-5205	ABOLISHED	--	0.00%	0.00%	0.00%	Form 27D	Consumer Electronic	Exempted - Former 206C(94) removed by 2025 Act	Wed, 01 Apr 2020 00:00:00 GMT
GL-5201	394	--	2.00%	0.00%	0.00%	Form 27D	Scrap Yard Operator	Industrial waste processing clearing sales	Wed, 01 Apr 2020 00:00:00 GMT
GL-5205	ABOLISHED	--	0.00%	0.00%	0.00%	Form 27D	Automotive Vendor	Exempted - Former 206C(94) removed by 2025 Act	Wed, 01 Apr 2020 00:00:00 GMT
GL-5201	394	--	2.00%	0.00%	0.00%	Form 27D	Quarry Operations	Stone aggregate logistics trade processing	Wed, 01 Apr 2020 00:00:00 GMT
GL-5205	ABOLISHED	--	0.00%	0.00%	0.00%	Form 27D	Textile Manufacturer	Exempted - Former 206C(94) removed by 2025 Act	Wed, 01 Apr 2020 00:00:00 GMT

Showing 1-10 of 13 records

Figure 4: Form 26AS — TCS Records tab showing section codes with applicable rates

6.3 Books of Accounts

The Books of Accounts screen is the primary income data module. It displays income, TDS deductible, and reconciliation data pulled from the uploaded sales register or manual entry files for the selected financial year. The screen has five KPI cards and five data tabs.

KPI Cards

- Income as per Sales Register — total gross invoiced amount for the selected FY
- Adjustments — credit notes and deductions (if any)
- Linked Adjustments — amounts matched to specific invoices
- Total Income — net taxable income after adjustments
- TDS Deductible — expected TDS deduction amount

Summary Tab

The Summary tab shows a customer-wise aggregate view — Taxable Amount, Gross Amount, TDS Deductible, Collection Amount, and TDS Receivable. It can be filtered by customer status (All Customers / Active / Inactive).

Books of Accounts
Income, TDS deductible, and reconciliation data pulled from the sales register for the selected financial year.

FINANCIAL YEAR: FY 2023-24 | FY 2023-24 | Load

Category	Amount (₹)
INCOME AS PER SALES REGISTER (gross invoiced amount)	5,29,680.00
ADJUSTMENTS (credit notes / deductions)	0.00
LINKED ADJUSTMENTS (matched to invoices)	0.00
TOTAL INCOME (net taxable income)	4,83,880.00
TDS DEDUCTIBLE (expected deduction)	45,880.00

CUSTOMER NAME	TAXABLE AMOUNT	GROSS AMOUNT	TDS DEDUCTIBLE	COLLECTION AMOUNT	TDS RECEIVABLE
ALISHA	65,000.00	69,000.00	4,000.00	65,000.00	4,000.00
ASHRAFA	60,000.00	66,000.00	6,000.00	60,000.00	6,000.00
JAMIE	65,000.00	71,500.00	6,500.00	65,000.00	6,500.00
JAMIESON	65,000.00	71,500.00	6,500.00	65,000.00	6,500.00
NIKARIKA	60,000.00	74,000.00	6,000.00	60,000.00	6,000.00
RAVISHANKAR	70,000.00	77,000.00	7,000.00	70,000.00	7,000.00
SANJANA R	50,000.00	55,000.00	5,000.00	50,000.00	5,000.00
ZAINA	40,000.00	44,000.00	4,000.00	40,000.00	4,000.00
Total (8 customers)	4,83,880.00	5,29,680.00	45,880.00	4,83,880.00	45,880.00

Showing 1-8 of 8 customers

Figure 5: Books of Accounts — Summary tab (FY 2023-24) showing 8 customers with taxable and TDS amounts

Detailed Tab

The Detailed tab shows invoice-level data — Invoice No, Invoice Date, Taxable Amount, Gross Amount, TDS Deductible, Collection Amount, TDS Receivable, and Remarks. It supports month-level filtering (All Months / January through December) and free-text search by customer name or invoice number.

Books of Accounts
Income, TDS deductible, and reconciliation data pulled from the sales register for the selected financial year.

FINANCIAL YEAR: FY 2023-24 | FY 2023-24 | Load

CUSTOMER NAME	INVOICE NO	INVOICE DATE	TAXABLE AMOUNT	GROSS AMOUNT	TDS DEDUCTIBLE	COLLECTION AMOUNT	TDS RECEIVABLE	REMARKS
NIKARIKA	INV-2023-003	2023-07-31	30,000.00	33,000.00	3,000.00	30,000.00	3,000.00	Automated Sync
JAMIE	INV-2023-003	2023-07-31	25,000.00	27,500.00	2,500.00	25,000.00	2,500.00	Automated Sync
ZAINA	INV-2023-004	2023-07-31	10,000.00	11,000.00	1,000.00	10,000.00	1,000.00	Automated Sync
JAMIESON	INV-2023-004	2023-07-30	10,000.00	11,000.00	1,000.00	10,000.00	1,000.00	Automated Sync
JAMIE	INV-2023-002	2023-07-30	20,000.00	22,000.00	2,000.00	20,000.00	2,000.00	Automated Sync
ZAINA	INV-2023-003	2023-07-30	10,000.00	11,000.00	1,000.00	10,000.00	1,000.00	Automated Sync
JAMIESON	INV-2023-003	2023-07-29	10,000.00	11,000.00	1,000.00	10,000.00	1,000.00	Automated Sync
JAMIE	INV-2023-005	2023-07-29	20,000.00	22,000.00	2,000.00	20,000.00	2,000.00	Automated Sync
ZAINA	INV-2023-002	2023-07-29	10,000.00	11,000.00	1,000.00	10,000.00	1,000.00	Automated Sync
JAMIESON	INV-2023-002	2023-07-28	20,000.00	22,000.00	2,000.00	20,000.00	2,000.00	Automated Sync

Showing 1-10 of 29 entries

Figure 6: Books of Accounts — Detailed tab showing invoice-level records with month filter

Books of Accounts
Income, TDS deductible, and reconciliation data pulled from the sales register for the selected financial year.

FINANCIAL YEAR: FY 2023-24

Summary | Detailed | Income Records | Unreferenced Entries | Upload History

Search by customer, invoice no, description...

CUSTOMER NAME	INVOICE NO	INVOICE DATE	TAXABLE AMOUNT	GROSS AMOUNT	TDS DEDUCTIBLE	COLLECTION AMOUNT	TDS RECEIVABLE	REMARKS
NIKARIKA	INV-2023-003	2023-07-31	₹ 30,000.00	₹ 33,000.00	₹ 3,000.00	₹ 30,000.00	₹ 3,000.00	Auto
JAMIE	INV-2023-003	2023-07-31	₹ 25,000.00	₹ 27,500.00	₹ 2,500.00	₹ 25,000.00	₹ 2,500.00	Auto
ZAINA	INV-2023-004	2023-07-31	₹ 10,000.00	₹ 11,000.00	₹ 1,000.00	₹ 10,000.00	₹ 1,000.00	Auto
JAMESON	INV-2023-004	2023-07-30	₹ 10,000.00	₹ 11,000.00	₹ 1,000.00	₹ 10,000.00	₹ 1,000.00	Auto
JAMIE	INV-2023-002	2023-07-30	₹ 20,000.00	₹ 22,000.00	₹ 2,000.00	₹ 20,000.00	₹ 2,000.00	Auto
ZAINA	INV-2023-003	2023-07-30	₹ 10,000.00	₹ 11,000.00	₹ 1,000.00	₹ 10,000.00	₹ 1,000.00	Auto
JAMESON	INV-2023-003	2023-07-29	₹ 10,000.00	₹ 11,000.00	₹ 1,000.00	₹ 10,000.00	₹ 1,000.00	Auto
JAMIE	INV-2023-005	2023-07-29	₹ 20,000.00	₹ 22,000.00	₹ 2,000.00	₹ 20,000.00	₹ 2,000.00	Auto
ZAINA	INV-2023-002	2023-07-29	₹ 10,000.00	₹ 11,000.00	₹ 1,000.00	₹ 10,000.00	₹ 1,000.00	Auto
JAMESON	INV-2023-002	2023-07-28	₹ 20,000.00	₹ 22,000.00	₹ 2,000.00	₹ 20,000.00	₹ 2,000.00	Automated Sync

Showing 1-10 of 29 entries

Figure 7: Books of Accounts — Detailed tab with month dropdown showing all calendar months

Income Records Tab

The Income Records tab provides row-level access to the underlying `income_records` table. It shows Customer Name, Document No, Document Date, Document Type, Taxable Amount, Tax Amount, Gross Amount, Source (Sales Register / Manual Entry), and Upload Reference. The Source filter (All Sources / Sales Register / Manual Entry) is auto-detected from the uploaded file type — TXT files are classified as Sales Register, XLSX files without an explicit source column are classified as Manual Entry.

Books of Accounts
Income, TDS deductible, and reconciliation data pulled from the sales register for the selected financial year.

FINANCIAL YEAR: FY 2023-24

Summary | Detailed | Income Records | Unreferenced Entries | Upload History

Search by customer, invoice no, description...

CUSTOMER NAME	DOCUMENT NO	DOCUMENT DATE	DOCUMENT TYPE	TAXABLE AMOUNT	TAX AMOUNT	GROSS AMOUNT	SOURCE	UPLOAD REF
ALISHA	INV-2023-003	2023-04-16	Invoice	₹ 25,000.00	₹ 1,500.00	₹ 26,500.00	sales_register	BOOKS-DA6P6F37
ALISHA	INV-2023-002	2023-04-16	Invoice	₹ 20,000.00	₹ 1,000.00	₹ 21,000.00	sales_register	BOOKS-DA6P6F37
ALISHA	INV-2023-001	2023-04-15	Invoice	₹ 20,000.00	₹ 1,500.00	₹ 21,500.00	sales_register	BOOKS-DA6P6F37
ASHRAFA	INV-2023-002	2023-04-16	Invoice	₹ 10,000.00	₹ 1,000.00	₹ 11,000.00	sales_register	BOOKS-DA6P6F37
ASHRAFA	INV-2023-001	2023-04-15	Invoice	₹ 20,000.00	₹ 2,000.00	₹ 22,000.00	sales_register	BOOKS-DA6P6F37
ASHRAFA	INV-2023-003	2023-04-15	Invoice	₹ 20,000.00	₹ 2,000.00	₹ 22,000.00	sales_register	BOOKS-DA6P6F37
ASHRAFA	INV-2023-004	2023-04-15	Invoice	₹ 10,000.00	₹ 1,000.00	₹ 11,000.00	sales_register	BOOKS-DA6P6F37
JAMIE	INV-2023-003	2023-07-31	Invoice	₹ 25,000.00	₹ 2,500.00	₹ 27,500.00	sales_register	BOOKS-DA6P6F37
JAMIE	INV-2023-002	2023-07-30	Invoice	₹ 20,000.00	₹ 2,000.00	₹ 22,000.00	sales_register	BOOKS-DA6P6F37
JAMIE	INV-2023-001	2023-07-29	Invoice	₹ 20,000.00	₹ 2,000.00	₹ 22,000.00	sales_register	BOOKS-DA6P6F37

Showing 1-10 of 29 records

Figure 8: Books of Accounts — Income Records tab showing per-invoice detail with Source and Upload Ref columns

Books of Accounts
Income, TDS deductible, and reconciliation data pulled from the sales register for the selected financial year.

FINANCIAL YEAR: FY 2023-24

Summary | **Detailed** | **Income Records** | **Unreferenced Entries** | **Upload History**

INCOME AS PER SALES REGISTER
₹ 5,29,680.00
gross taxable amount

ADJUSTMENTS
₹ 0.00
credit notes / deductions

LINKED ADJUSTMENTS
₹ 0.00
matched to invoices

TOTAL INCOME
₹ 4,83,880.00
net taxable income

TDS DEDUCTIBLE
₹ 45,880.00
expected deduction

Income Records

CUSTOMER	DATE	DOCUMENT	AMOUNT	TAXABLE INCOME	SOURCE
ALISHA	2023-04-15	INVOICE	₹ 20,000.00	₹ 1,000.00	sales_register
ALISHA	2023-04-15	INVOICE	₹ 20,000.00	₹ 1,000.00	sales_register
ASHIAFA	2023-04-15	INVOICE	₹ 10,000.00	₹ 1,000.00	sales_register
ASHIAFA	2023-04-15	INVOICE	₹ 20,000.00	₹ 2,000.00	sales_register
ASHIAFA	2023-04-15	INVOICE	₹ 20,000.00	₹ 2,000.00	sales_register
ASHIAFA	2023-04-15	INVOICE	₹ 10,000.00	₹ 1,000.00	sales_register
JAMIE	2023-07-31	INVOICE	₹ 25,000.00	₹ 2,500.00	sales_register
JAMIE	2023-07-30	INVOICE	₹ 20,000.00	₹ 2,000.00	sales_register
JAMIE	2023-07-29	INVOICE	₹ 20,000.00	₹ 2,000.00	sales_register

Showing 1-10 of 29 records

Figure 9: Books of Accounts — Income Records with All Sources filter showing Sales Register and Manual Entry options

Unreferenced Entries Tab

The Unreferenced Entries tab captures any income records that could not be matched to a customer in the Customer Master. It categorizes unmatched reasons as: No PAN Match, No TAN Match, No Customer Match, or Possible Duplicate. A green tick and 'No unreferenced entries — all records matched' message confirms clean data integrity when all records are successfully linked.

Books of Accounts
Income, TDS deductible, and reconciliation data pulled from the sales register for the selected financial year.

FINANCIAL YEAR: FY 2023-24

Summary | **Detailed** | **Income Records** | **Unreferenced Entries** | **Upload History**

INCOME AS PER SALES REGISTER
₹ 5,29,680.00
gross taxable amount

ADJUSTMENTS
₹ 0.00
credit notes / deductions

LINKED ADJUSTMENTS
₹ 0.00
matched to invoices

TOTAL INCOME
₹ 4,83,880.00
net taxable income

TDS DEDUCTIBLE
₹ 45,880.00
expected deduction

Unreferenced Entries

Search unreferenced entries...

RAW NAME / IDENTIFIER	DOCUMENT NO	DOCUMENT DATE	AMOUNT	UNREF REASON	UPLOAD REF	REMARKS
No unreferenced entries — all records matched						

Showing 0-0 of 0 entries

Figure 10: Books of Accounts — Unreferenced Entries tab showing zero unmatched records

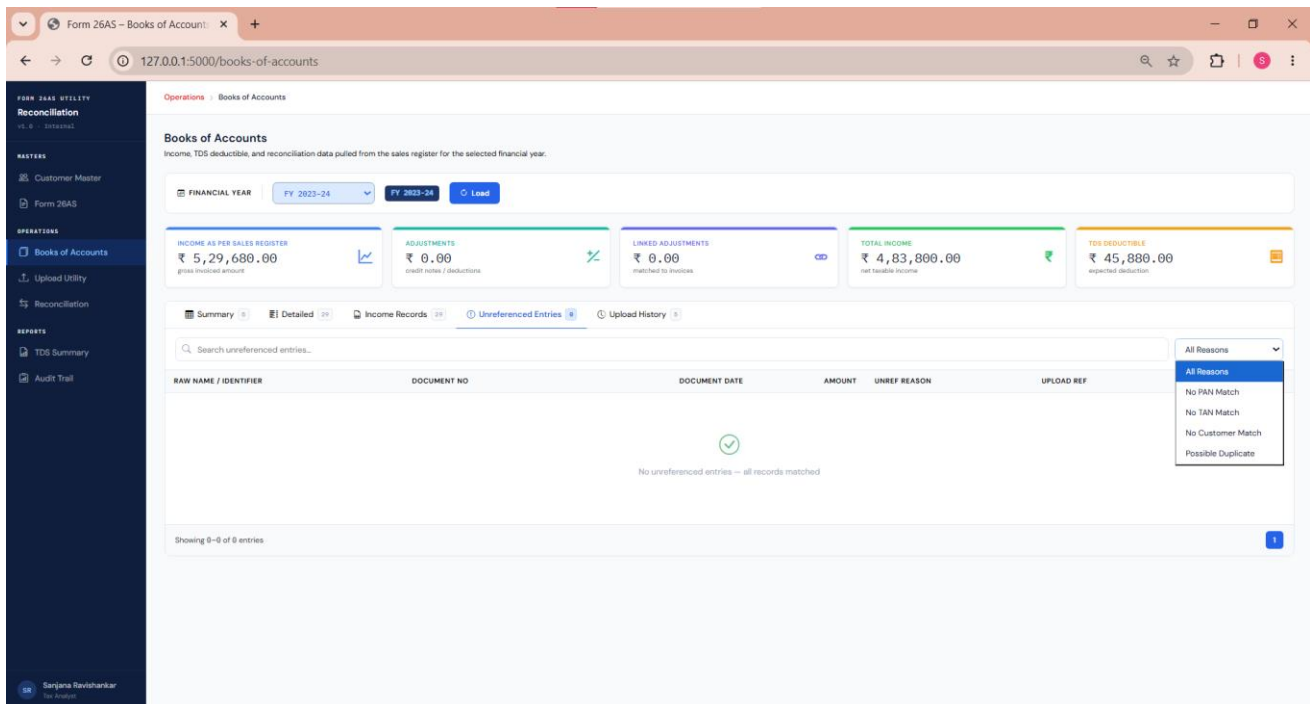


Figure 11: Books of Accounts — Unreferenced Entries filter showing reason categories

Upload History Tab

The Upload History tab shows a log of all Books of Accounts file uploads, with filename, upload timestamp, file size, and status badge (success/failed).

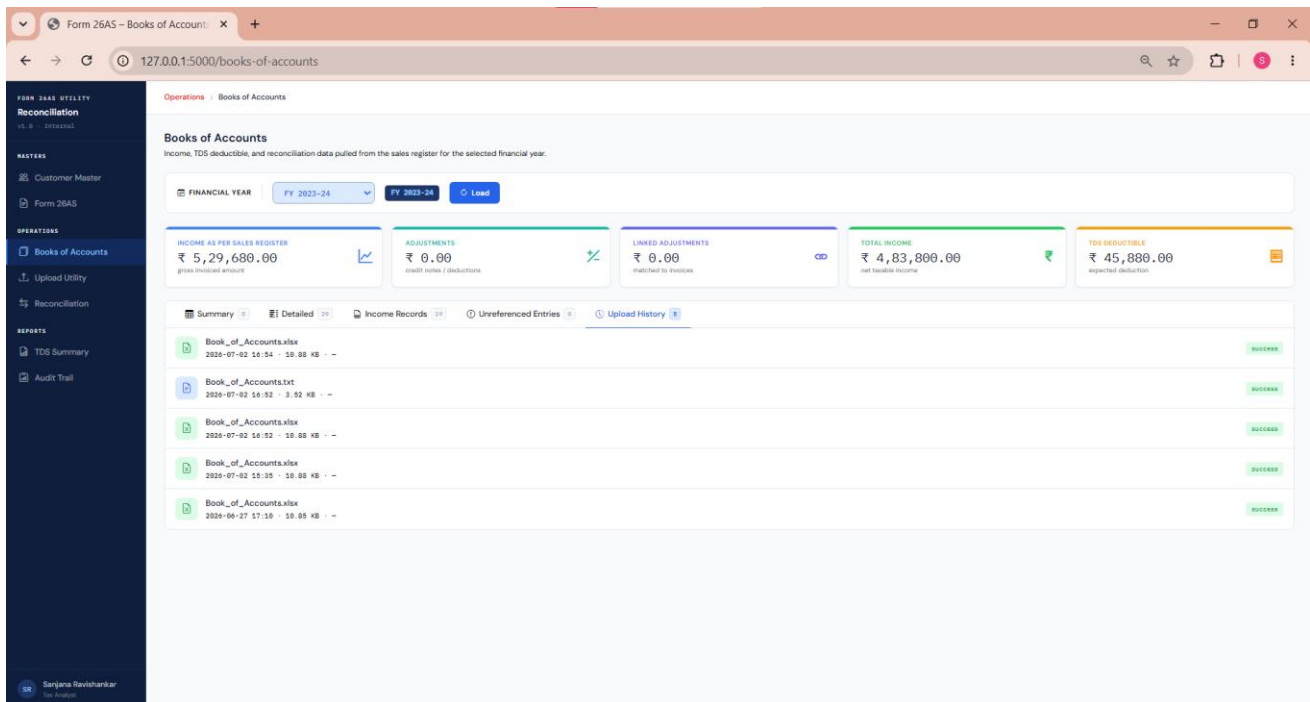


Figure 12: Books of Accounts — Upload History tab showing the uploaded Books file

6.4 Upload Utility

The Upload Utility is the central file ingestion module. It supports three upload types — Customer Master (Excel), Books of Accounts (Excel/TXT), and Form 26AS (TDS/TCS Credits via PDF/XLSX/TXT). Users select the upload type from a dropdown, then drag and drop or click to browse for the file.

Key features of the Upload Utility include:

- Template downloads — Excel and TXT templates are provided for Customer Master and Books of Accounts upload type with pre-defined column headers
- Expected columns preview — shown below the upload type selector to guide users on file format
- SHA-256 hash deduplication — identical file uploads are detected and blocked with an informative message
- Per-row financial year detection — each row's document date is used to determine its correct FY, enabling multi-FY files to be handled correctly
- Upsert logic — re-uploading a file with corrections automatically updates existing records without creating duplicates
- FY date validation — rows with dates outside the detected financial year range are flagged in the upload result
- Upload history log — all uploads are recorded with module, FY, quarter, file size, row count, status, and timestamp

The screenshot displays the 'Form 26AS - Upload Utility' web interface. The left sidebar contains navigation links for 'MASTERS' (Customer Master, Form 26AS), 'OPERATIONS' (Books of Accounts, Upload Utility), and 'REPORTS' (TDS Summary, Audit Trail). The main content area is titled 'Upload Utility' and includes instructions: 'Upload customer data, books of accounts, or Form 26AS files. Select the upload type before choosing a file.' Below this, there is an 'Upload New File' section with a dropdown menu for 'UPLOAD TYPE' set to 'Customer Master (Excel)'. A link for 'Download Excel Template' is visible. Below the dropdown, the 'Expected Columns' are listed: 'Customer Code | Customer Name | PAN Number | DOB | Status | Contact'. A large area with a cloud icon and text 'Drag & drop your file here or click to browse' is provided for file upload. Below this is an 'Upload File' button. At the bottom, there is a table titled 'All Uploads' with columns: FILE NAME, MODULE, FY, QUARTER, SIZE, STATUS, ROWS, UPLOADED BY, and UPLOADED AT. The table lists several uploads, including 'Book_of_Accounts.xlsx', 'customer_master.xlsx', and various Form 26AS PDF files, all with a status of 'SUCCESS'.

FILE NAME	MODULE	FY	QUARTER	SIZE	STATUS	ROWS	UPLOADED BY	UPLOADED AT
Book_of_Accounts.xlsx	books_of_accounts	2023-24	Q1	55.55 KB	SUCCESS	5	—	2024-06-27 17:10
customer_master.xlsx	customer_master	N/A	NA	9.39 KB	SUCCESS	55	—	2024-06-27 17:08
customer_master.xlsx	customer_master	N/A	NA	9.39 KB	SUCCESS	55	—	2024-06-27 17:02
QWYLR3954P-2021.pdf	form26as	2020-21	Q2	277.48 KB	SUCCESS	5	—	2024-06-27 16:50
XYREC8324U-2024.pdf	form26as	2023-24	Q2	277.54 KB	SUCCESS	5	—	2024-06-27 14:25
TRWUE7653Z-2024.pdf	form26as	2023-24	Q2	277.15 KB	SUCCESS	5	—	2024-06-27 14:28
TRQUE7653Z-2024.pdf	form26as	2023-24	Q2	277.15 KB	SUCCESS	5	—	2024-06-27 14:28

Figure 13: Upload Utility — Customer Master upload type with Excel template download

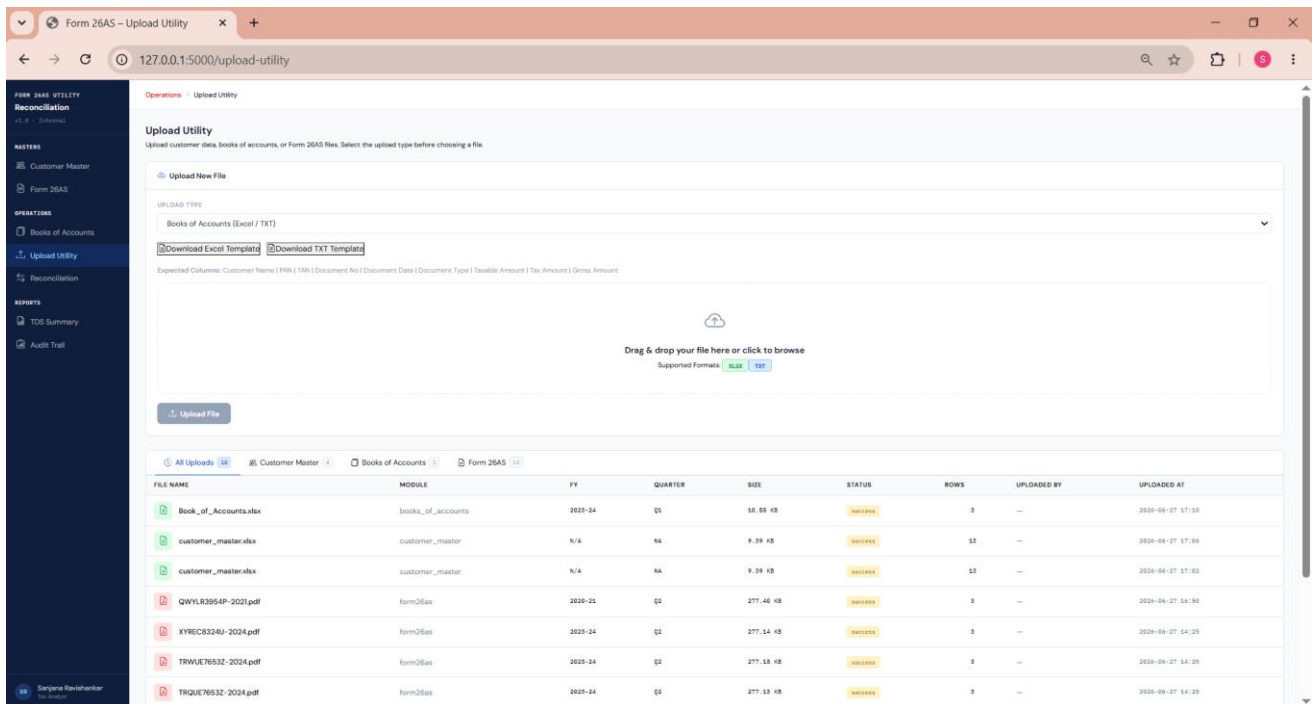


Figure 14: Upload Utility — Books of Accounts upload type showing Excel and TXT template options

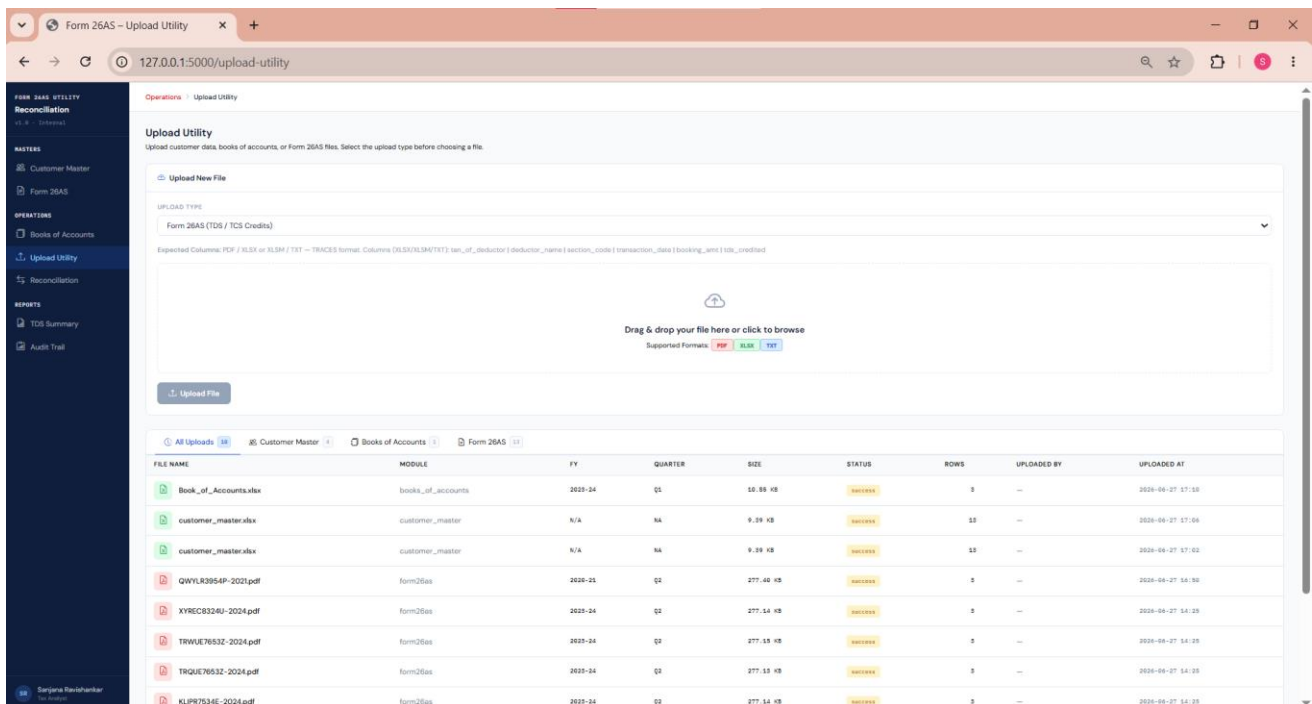


Figure 15: Upload Utility — Form 26AS upload type supporting PDF, XLSX, and TXT formats with full upload history

6.5 Reconciliation

The Reconciliation screen is the core operational module of the system. It matches Books of Accounts entries against Form 26AS credits for each customer and financial year, computing variance amounts and match scores, and displaying categorized reconciliation statuses.

KPI Cards

- System TDS — total TDS deductible as per Books of Accounts
- Form 26AS Credit — total TDS credited as per uploaded TRACES PDFs
- Book Credit — total gross collections received
- % Credit — ratio of 26AS Credit to System TDS, expressed as a percentage

Status Tabs

Records are automatically classified into the following statuses based on the match score:

- Reconciled — 26AS Credit equals System TDS (variance < ₹1.00), match score 100%
- Likely Match — match score ≥ 85%, small variance present
- Suggested Match — match score ≥ 50%, larger variance
- Open — no 26AS data uploaded for this customer, or match score < 50%
- Reserved / Not Liable / Disabled — manual status overrides available via the Change action

The screenshot displays the 'Form 26AS - Reconciliation' interface. At the top, there's a navigation bar with 'Form 26AS - Reconciliation' and a search bar. Below this, a sidebar on the left contains links for 'MAINTENANCE' (Customer Master, Form 26AS), 'OPERATIONS' (Books of Accounts, Upload Utility, Reconciliation), and 'REPORTS' (TDS Summary, Audit Trail). The main area is titled 'Reconciliation' and shows a summary of the reconciliation process for the financial year 2022-23. It includes four KPI cards: SYSTEM TDS (₹ 48,380.00), FORM 26AS CREDIT (₹ 46,880.00), BOOK CREDIT (₹ 5,32,180.00), and % CREDIT (96.9%). Below these cards, there are tabs for 'Grand Total', 'Open', 'Reconciled', 'Likely Match', 'Suggested', 'Reserved', 'Not Liable', and 'Disabled'. The 'Grand Total' tab is selected, showing a table of reconciliation records. The table has columns for CUSTOMER, PAN, TAN, SYSTEM TDS, 26AS CREDIT, BOOK CREDIT, VARIANCE, MATCH SCORE, STATUS, and REMARKS. A dropdown menu on the right side of the table allows filtering by 'All Variances', 'Exact Match', 'Over-credited', and 'Under-credited'. The table shows several records with their respective match scores and statuses, and a 'Change' button for each record. The bottom of the table shows a 'Total (B)' row with the grand totals for each column.

CUSTOMER	PAN	TAN	SYSTEM TDS	26AS CREDIT	BOOK CREDIT	VARIANCE	MATCH SCORE	STATUS	REMARKS
ALISHA	KLH4784E	WVC4883F	₹ 6,500.00	₹ 6,500.00	₹ 71,500.00	₹ 500.00	92%	Likely_Match	Automated engine validation complete.
AISHRAFA	KLH4784E	L4V7638E	₹ 6,000.00	₹ 6,000.00	₹ 66,000.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.
JAME	TN4U785Z	HPV1848E	₹ 6,500.00	₹ 6,500.00	₹ 71,500.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.
JAMESON	TN4U785Z	KLVC848E	₹ 6,500.00	₹ 6,500.00	₹ 71,500.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.
NIKARKA	KZC4974Q	DNV73724X	₹ 6,000.00	₹ 5,000.00	₹ 74,000.00	₹ 1,000.00	85%	Likely_Match	Automated engine validation complete.
RAVISHANKAR	ZJL47834V	P4V7638E	₹ 7,000.00	₹ 7,000.00	₹ 77,000.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.
SANJANA R	PPY4895Z	SLR6233E	₹ 5,000.00	₹ 5,000.00	₹ 55,000.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.
ZAINA	XYR6C524U	W7V75485X	₹ 4,000.00	₹ 4,000.00	₹ 44,000.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.
Total (B)			₹ 48,380.00	₹ 46,880.00	₹ 5,32,180.00	₹ 1,500.00			

Figure 16: Reconciliation — Grand Total view with variance filter (All Variances / Exact Match / Over-credited / Under-credited)

Form 26AS – Reconciliation

127.0.0.1:5000/reconciliation

Operations > Reconciliation

Reconciliation

Match book entries against Form 26AS credits. Review and confirm each reconciliation status.

FINANCIAL YEAR: FY 2023-24

Search customer, PAN, TAN: All Customers

Load

SYSTEM TDS: ₹ 48,380.00

FORM 26AS CREDIT: ₹ 46,880.00

BOOK CREDIT: ₹ 5,32,180.00

% CREDIT: 96.9%

Grand Total: Open Reconciled Likely Match Suggested Reserved Not Liable Disabled

All Records

CUSTOMER	PAN	TAN	SYSTEM TDS	26AS CREDIT	BOOK CREDIT	VARIANCE	MATCH SCORE	STATUS	REMARKS	ACTION
ADIRAFI	KLZPT532E	LVPT7623E	₹ 6,000.00	₹ 6,000.00	₹ 60,000.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.	Change
JAME	TRQU7453Z	KPVC8483E	₹ 6,500.00	₹ 6,500.00	₹ 71,500.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.	Change
JAMESON	TRQU7453Z	KLVC8483E	₹ 6,500.00	₹ 6,500.00	₹ 71,500.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.	Change
RAVISHANKAR	23LPT763W	PVPT7633E	₹ 7,000.00	₹ 7,000.00	₹ 77,000.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.	Change
SANJANA R	PFFR8954Z	BLPM2333E	₹ 5,800.00	₹ 5,800.00	₹ 55,800.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.	Change
ZAINA	XVRC8324U	WST18248X	₹ 4,000.00	₹ 4,000.00	₹ 44,000.00	₹ 0.00	100%	Reconciled	Automated engine validation complete.	Change
Total (6)			₹ 35,800.00	₹ 35,800.00	₹ 3,85,800.00	₹ 0.00				

Showing 1-6 of 6 records

Figure 17: Reconciliation — Reconciled tab showing 6 customers with 100% match score and zero variance

Form 26AS – Reconciliation

127.0.0.1:5000/reconciliation

Operations > Reconciliation

Reconciliation

Match book entries against Form 26AS credits. Review and confirm each reconciliation status.

FINANCIAL YEAR: FY 2023-24

Search customer, PAN, TAN: All Customers

Load

SYSTEM TDS: ₹ 48,380.00

FORM 26AS CREDIT: ₹ 46,880.00

BOOK CREDIT: ₹ 5,32,180.00

% CREDIT: 96.9%

Grand Total: Open Reconciled Likely Match Suggested Reserved Not Liable Disabled

All Records

CUSTOMER	PAN	TAN	SYSTEM TDS	26AS CREDIT	BOOK CREDIT	VARIANCE	MATCH SCORE	STATUS	REMARKS	ACTION
ALISHA	KLH8784E	KHVS4453P	₹ 4,500.00	₹ 4,000.00	₹ 71,500.00	₹ 500.00	92%	Likely Match	Automated engine validation complete.	Change
NIKARIKA	KZCV1974Q	CVPT5724X	₹ 6,000.00	₹ 5,000.00	₹ 74,000.00	₹ 1,000.00	85%	Likely Match	Automated engine validation complete.	Change
Total (2)			₹ 13,300.00	₹ 11,000.00	₹ 1,45,500.00	₹ 1,500.00				

Showing 1-2 of 2 records

Figure 18: Reconciliation — Likely Match tab showing ALISHA (92%) and NIKARIKA (85%) with small variances

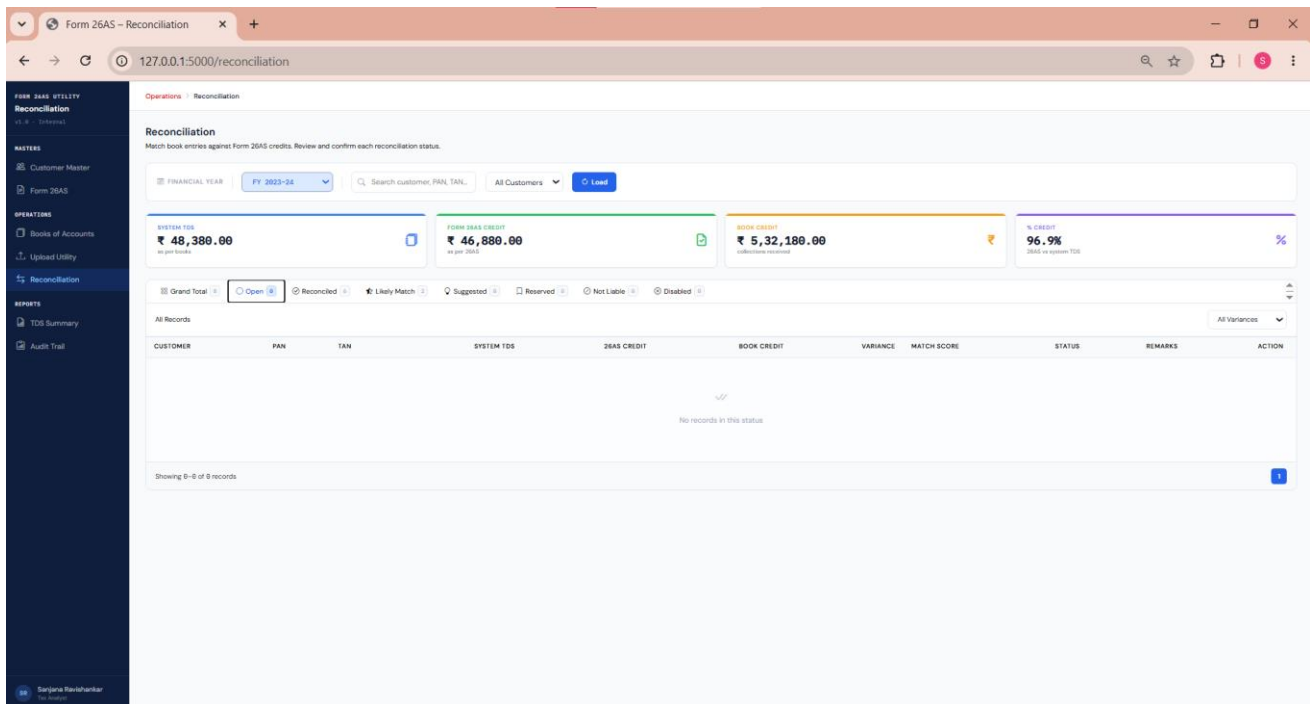


Figure 19: Reconciliation — Open tab showing no records (all customers fully reconciled or in likely match)

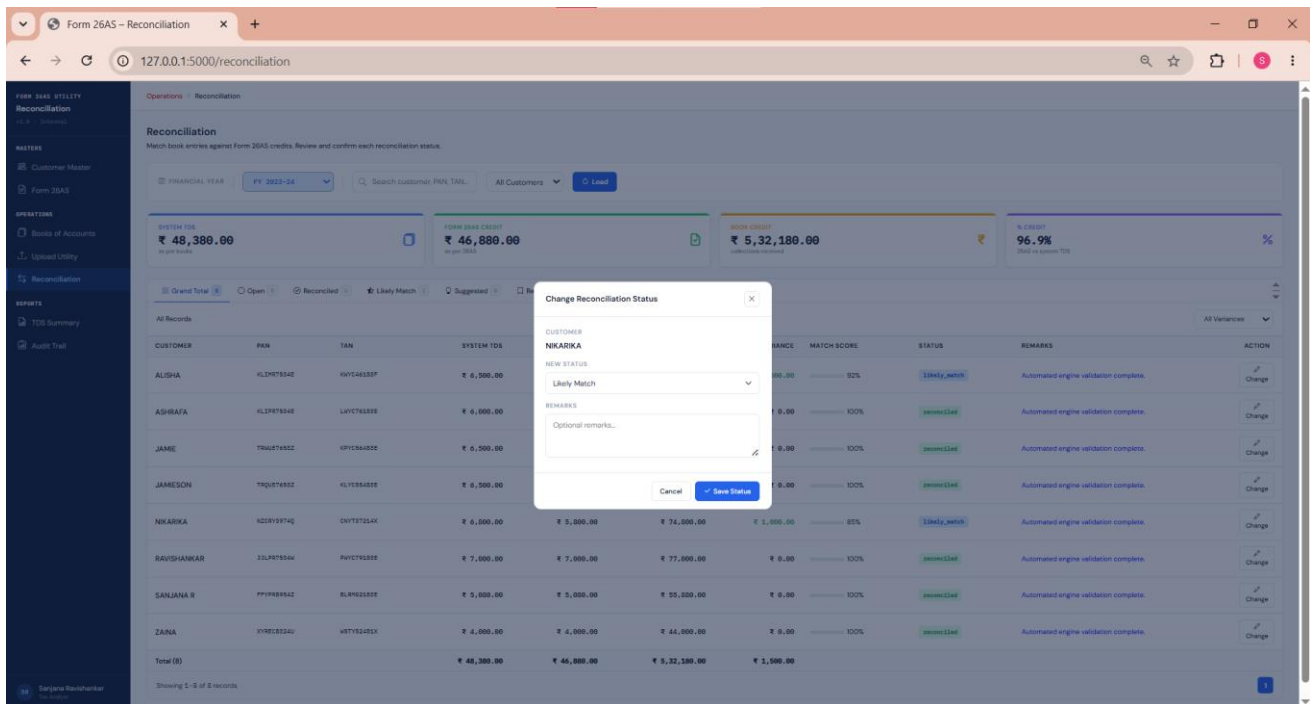


Figure 20: Reconciliation — Dialog box showing “Change Reconciliation Status” along with the provision to give optional remarks

6.6 TDS Summary Report

The TDS Summary Report provides a consolidated view of TDS deductible versus Form 26AS credits per customer and TDS section. It can be filtered by Financial Year, Customer, and Section. Four summary KPI cards show Total Taxable Amount, TDS Deductible, Form 26AS Credit, and Net Variance.

The table shows each customer's PAN, TAN, Section code, Taxable Amount, TDS Deductible, 26AS Credit, Variance (red for positive, green for zero/negative), and Invoice count. The report can be exported as a summary Excel file using the Export Summary Report button.

The financial year filter allows viewing the same customer base across multiple FYs — for example, customers with invoice dates in FY 2022-23 appear separately from FY 2023-24, enabling multi-year comparison without data contamination.

CUSTOMER	PAN	TAN	SECTION	TAXABLE AMT	TDS DEDUCTIBLE	26AS CREDIT	VARIANCE	INVOICES
ALISHA	KLME7834E	HUY44815P	192	₹ 65,000.00	₹ 6,500.00	₹ 6,000.00	₹ 500.00	3
ADHIRAJ	KLDP7834E	LUPY7615B	192	₹ 60,000.00	₹ 6,000.00	₹ 6,000.00	₹ 0.00	4
JAME	TNAUT785Z	KPYC8485E	192	₹ 65,000.00	₹ 6,500.00	₹ 6,500.00	₹ 0.00	3
JAMESON	TQQU748Z	KLYC5485E	192	₹ 65,000.00	₹ 6,500.00	₹ 6,500.00	₹ 0.00	4
NIKARIKA	KZCR1974Q	CMT73724X	192	₹ 60,000.00	₹ 6,000.00	₹ 5,000.00	₹ 1,000.00	3
RAUSHANKAR	JJLP7834E	PWY7915B	192	₹ 70,000.00	₹ 7,000.00	₹ 7,000.00	₹ 0.00	3
SANJANA R	FFYP8934Z	BLRQ2115E	192	₹ 50,000.00	₹ 5,000.00	₹ 5,000.00	₹ 0.00	5
ZAINA	XVBE1834U	WET92481Y	192	₹ 60,000.00	₹ 6,000.00	₹ 4,000.00	₹ 2,000.00	4
Total (8 rows)				₹ 4,83,880.00	₹ 46,380.00	₹ 46,580.00	₹ 1,500.00	

Figure 21: TDS Summary — FY 2023-24 view showing all customers with section codes, TDS amounts, and 26AS credits

CUSTOMER	PAN	TAN	SECTION	TAXABLE AMT	TDS DEDUCTIBLE	26AS CREDIT	VARIANCE
ALISHA	KL2P87834E	KWYC48150P	192	₹ 65,000.00	₹ 6,500.00	₹ 6,500.00	₹ 0.00
ADHRAFA	KL2P87834E	LWYC74153E	192	₹ 68,000.00	₹ 6,800.00	₹ 6,800.00	₹ 0.00
JAMEE	TN4L87832E	KPYC64832E	192	₹ 65,000.00	₹ 6,500.00	₹ 6,500.00	₹ 0.00
JAMESON	TG4U87832E	KLYC85483E	192	₹ 65,000.00	₹ 6,500.00	₹ 6,500.00	₹ 0.00
NIKABKA	NZ8R19374Q	DNV737214K	192	₹ 68,000.00	₹ 6,800.00	₹ 5,000.00	₹ 1,800.00
RAVISHANKAR	31L2P87834E	RWYC74153E	192	₹ 78,000.00	₹ 7,800.00	₹ 7,000.00	₹ 800.00
SANJANA R	PPYP88934E	BLRY62183E	192	₹ 98,000.00	₹ 9,800.00	₹ 9,800.00	₹ 0.00
ZARNA	XY8E19334U	WETV92483X	192	₹ 40,000.00	₹ 4,000.00	₹ 4,000.00	₹ 0.00
Total (8 rows)				₹ 4,83,888.00	₹ 46,388.00	₹ 46,888.00	₹ 1,500.00

Figure 22: TDS Summary — FY 2023-24 view showing all customers with filters – “Fully Matched”, “Under-credited”, “Over-credited” based on the calculated Variance

6.7 Audit Trail

The Audit Trail module provides a complete chronological log of all significant system events. Every file upload, reconciliation status change, and manual override is recorded with the event type, description, affected customer/file, user, and timestamp.

This module ensures full traceability and accountability — critical for a financial system where data accuracy and change history must be preserved for regulatory and audit purposes.

Form 26AS - Audit Trail

127.0.0.1:5000/audit-trail

Reports / Audit Trail

Complete log of all user actions across every module. This is a permanent, read-only record.

From dd-mm-yyyy To dd-mm-yyyy Action All Actions Module All Modules Severity All Apply Reset

TOTAL LOGS 25 (all recorded actions)

INFO 25 (no action)

WARNING 0 (requires attention)

CRITICAL 0 (high-priority events)

Search description, user, module...

#	TIMESTAMP	USER	ACTION	MODULE	DESCRIPTION	SEVERITY	IP ADDRESS
155	2020-04-27 17:10:55	System	UPLOAD	BOOKS OF ACCOUNTS	Books upload FY 2023-24: 3 inserted, 0 unmatched, 42 skipped.	INFO	127.0.0.1
156	2020-04-27 17:04:07	System	UPLOAD	CUSTOMER MASTER	Customer upload: 1 inserted, 12 updated, 0 skipped.	INFO	127.0.0.1
95	2020-04-27 17:02:51	System	UPLOAD	CUSTOMER MASTER	Customer upload: 1 inserted, 12 updated, 0 skipped.	INFO	127.0.0.1
96	2020-04-27 16:37:09	System	UPLOAD	BOOKS OF ACCOUNTS	Books upload FY 2023-24: 3 inserted, 0 unmatched, 42 skipped.	INFO	127.0.0.1
97	2020-04-27 16:30:55	System	UPLOAD	FORM26AS	Form 26AS upload FY 2020-21: 3 inserted, 0 skipped.	INFO	127.0.0.1
98	2020-04-27 15:05:16	System	UPLOAD	BOOKS OF ACCOUNTS	Books upload FY 2023-24: 13 inserted, 0 unmatched, 29 skipped.	INFO	127.0.0.1
99	2020-04-27 14:28:29	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
94	2020-04-27 14:28:17	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
93	2020-04-27 14:28:10	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
92	2020-04-27 14:28:04	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
91	2020-04-27 14:24:40	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
89	2020-04-27 14:13:07	System	UPLOAD	FORM26AS	Form 26AS upload FY 2022-23: 3 inserted, 0 skipped.	INFO	127.0.0.1
88	2020-04-27 14:12:59	System	UPLOAD	FORM26AS	Form 26AS upload FY 2022-23: 3 inserted, 0 skipped.	INFO	127.0.0.1
87	2020-04-27 14:12:52	System	UPLOAD	FORM26AS	Form 26AS upload FY 2022-23: 3 inserted, 0 skipped.	INFO	127.0.0.1

Showing 1-15 of 25 entries

Figure 23: Audit Trail — Full event log showing upload and reconciliation events with timestamps

Form 26AS - Audit Trail

127.0.0.1:5000/audit-trail

Reports / Audit Trail

Complete log of all user actions across every module. This is a permanent, read-only record.

From dd-mm-yyyy To dd-mm-yyyy Action All Actions Module All Modules Severity All Apply Reset

TOTAL LOGS 25 (all recorded actions)

INFO 25 (no action)

WARNING 0 (requires attention)

CRITICAL 0 (high-priority events)

Search description, user, module...

Login
 Logout
 Insert
 Update
 Delete
 Upload
 Export
 Execute

#	TIMESTAMP	USER	ACTION	MODULE	DESCRIPTION	SEVERITY	IP ADDRESS
155	2020-04-27 17:10:55	System	Export	BOOKS OF ACCOUNTS	Books upload FY 2023-24: 3 inserted, 0 unmatched, 42 skipped.	INFO	127.0.0.1
156	2020-04-27 17:04:07	System	Execute	CUSTOMER MASTER	Customer upload: 1 inserted, 12 updated, 0 skipped.	INFO	127.0.0.1
95	2020-04-27 17:02:51	System	UPLOAD	CUSTOMER MASTER	Customer upload: 1 inserted, 12 updated, 0 skipped.	INFO	127.0.0.1
96	2020-04-27 16:37:09	System	UPLOAD	BOOKS OF ACCOUNTS	Books upload FY 2023-24: 3 inserted, 0 unmatched, 42 skipped.	INFO	127.0.0.1
97	2020-04-27 16:30:55	System	UPLOAD	FORM26AS	Form 26AS upload FY 2020-21: 3 inserted, 0 skipped.	INFO	127.0.0.1
98	2020-04-27 15:05:16	System	UPLOAD	BOOKS OF ACCOUNTS	Books upload FY 2023-24: 13 inserted, 0 unmatched, 29 skipped.	INFO	127.0.0.1
99	2020-04-27 14:28:29	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
94	2020-04-27 14:28:17	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
93	2020-04-27 14:28:10	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
92	2020-04-27 14:28:04	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
91	2020-04-27 14:24:40	System	UPLOAD	FORM26AS	Form 26AS upload FY 2023-24: 3 inserted, 0 skipped.	INFO	127.0.0.1
89	2020-04-27 14:13:07	System	UPLOAD	FORM26AS	Form 26AS upload FY 2022-23: 3 inserted, 0 skipped.	INFO	127.0.0.1
88	2020-04-27 14:12:59	System	UPLOAD	FORM26AS	Form 26AS upload FY 2022-23: 3 inserted, 0 skipped.	INFO	127.0.0.1
87	2020-04-27 14:12:52	System	UPLOAD	FORM26AS	Form 26AS upload FY 2022-23: 3 inserted, 0 skipped.	INFO	127.0.0.1

Showing 1-15 of 25 entries

Figure 24: Audit Trail page showing “Action” filter

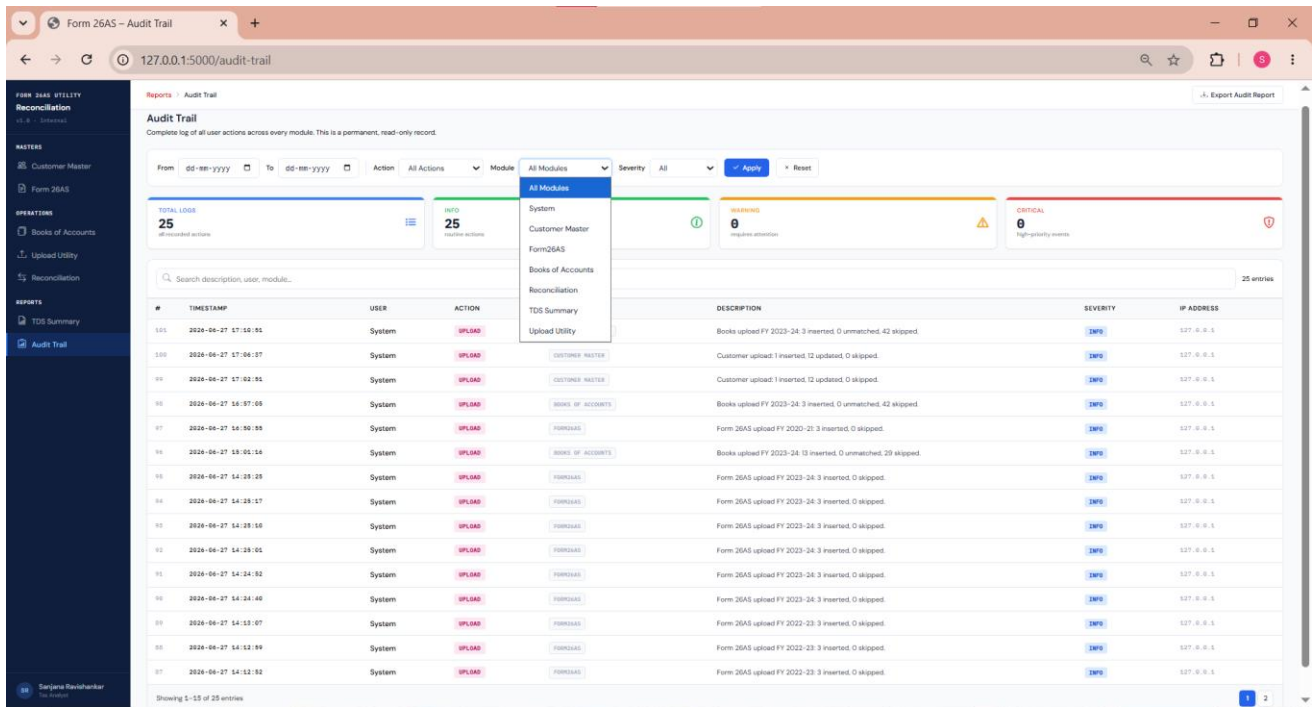


Figure 25: Audit Trail page showing “Module” filter

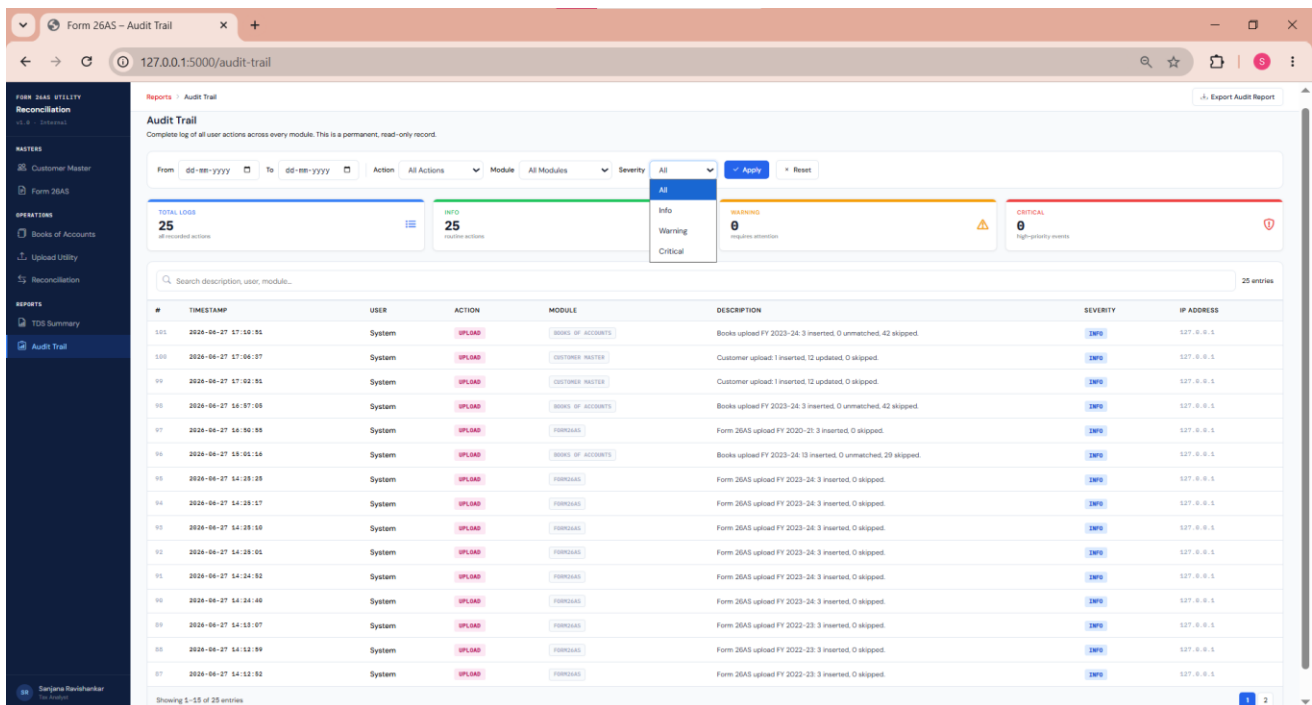


Figure 26: Audit Trail page showing “Severity” filter

6.8 Power BI Dashboard

A three-page Power BI Dashboard was built connected directly to the MySQL database via the ODBC connector. The dashboard uses a navy and gold financial theme consistent with the web application. Four MySQL views (**vw_tds_summary**, **vw_reconciliation**, **vw_invoice_detail**, **vw_upload_history**) were created specifically for Power BI consumption.

Page 1 — Executive Overview

The Overview page provides a high-level management view with 3 slicers (Financial year, Customer, Status), five Card visuals (Total Taxable, Total TDS Deductible, Total 26AS Credit, Net Variance, % Credit), a donut chart of reconciliation status breakdown, a quarterly invoice volume column chart, a clustered bar chart of TDS Deductible vs 26AS Credit by customer, and a Form 26AS vs Books of Accounts by Customer table with conditional formatting.

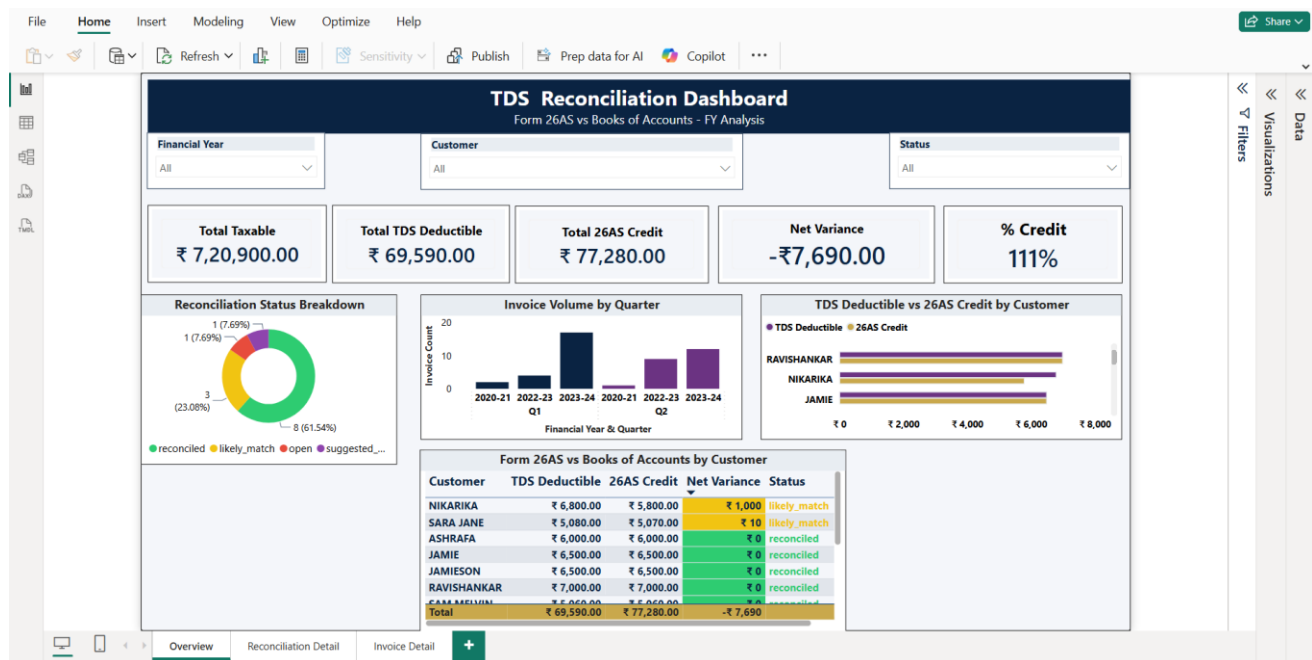


Figure 27: Power BI — Executive Overview page showing Cards, Reconciliation donut, Column Chart showing Invoice Volume by Quarter, TDS vs 26AS bar chart and a table showing Form 26AS vs Books of Accounts by Customer

Page 2 — Reconciliation Detail

The Reconciliation Detail page targets the finance and tax team. It shows a full customer-wise reconciliation table with System TDS, 26AS Credit, Variance (red/green/gold conditional formatting), Match Score in %, Status (with conditional formatting), and Remarks columns. A horizontal bar chart of Match Score % by customer includes a green threshold line at 85% to visually highlight customers approaching the likely match boundary. Drill-through is enabled on customer name.

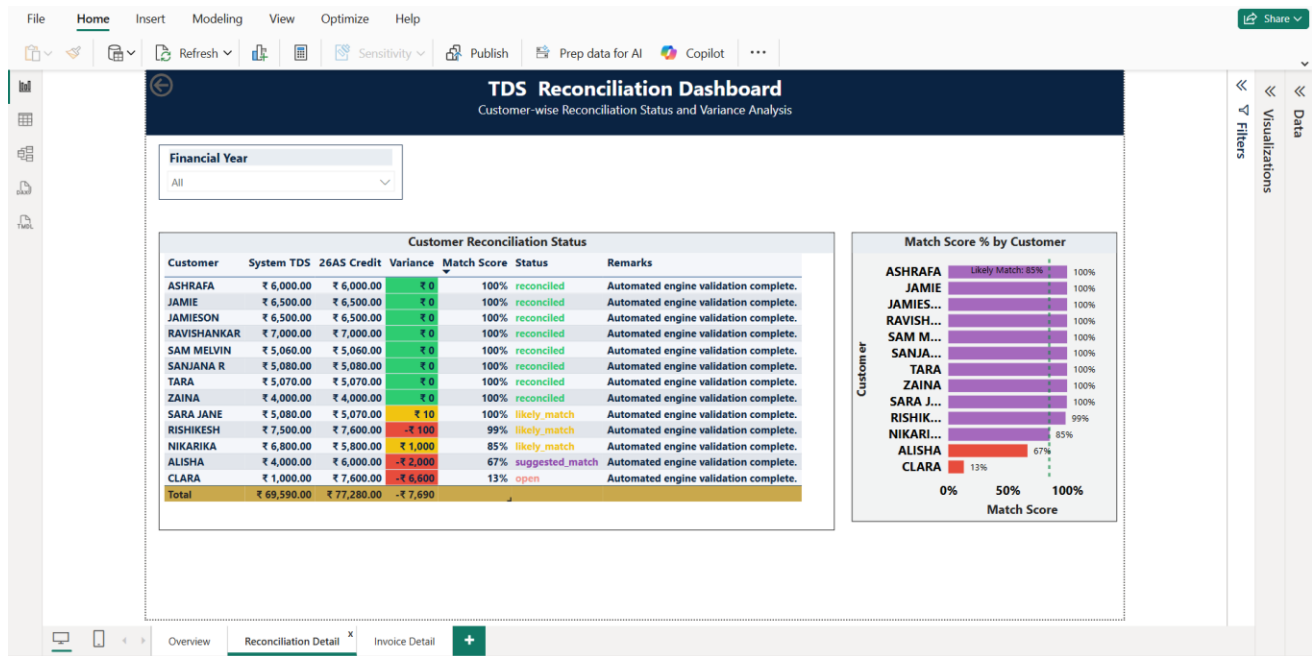


Figure 28: Power BI — Reconciliation Detail page with customer table showing conditional formatting on variance column and match score bar chart

Page 3 — Invoice Detail (Drill-through)

The Invoice Detail page is a drill-through target — accessed by right-clicking any customer in Page 2. It shows three summary cards (Customer Name, Total Invoices, Total Taxable Amount), a detailed invoice-level table (Document No, Document Date, Document Type, Taxable Amount, Tax Amount, Gross Amount, Entry Source, Quarter), and a monthly taxable amount trend column chart. A Back button returns the user to Page 2.

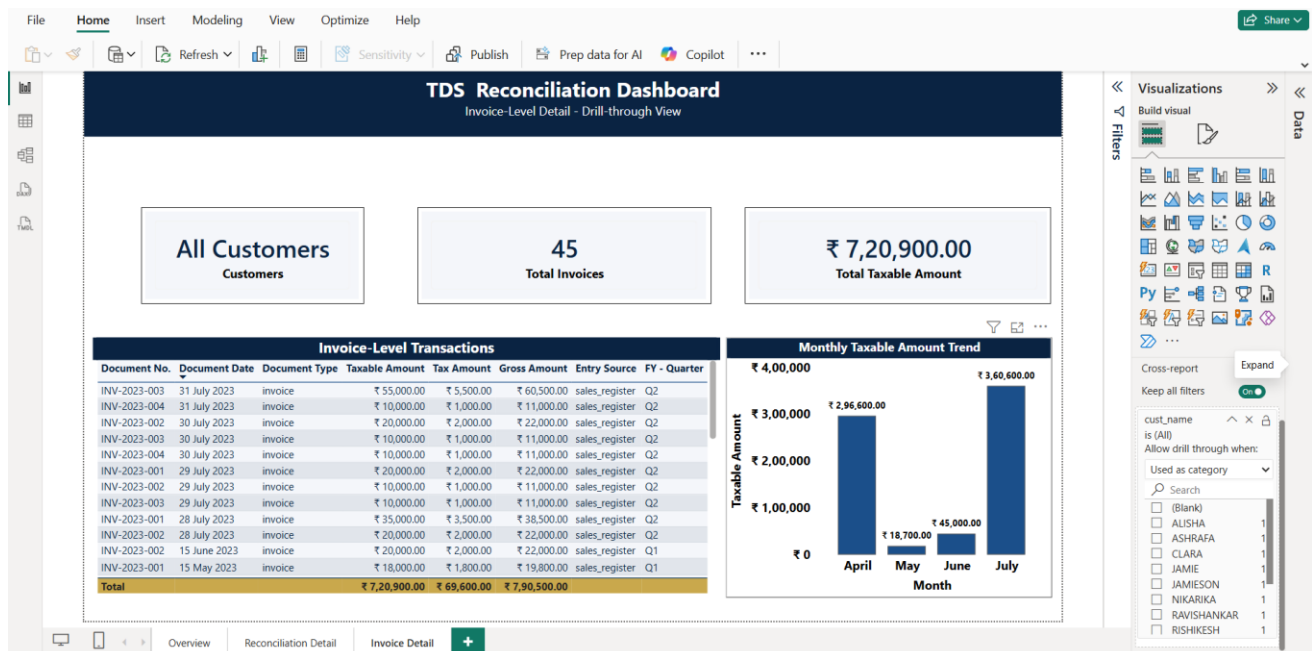


Figure 29: Power BI — Invoice Detail drill-through page showing invoice-level transactions and monthly trend

7. Key Technical Features

Intelligent PDF Parsing

The Form 26AS PDF parser uses pypdf for text-layer extraction and implements a specialized fix for a known pypdf issue where the TAN number is split mid-token (e.g., 'PWYC7918 3E' instead of 'PWYC79183E'). A targeted regex replacement collapses the split before TAN matching, ensuring all Form 26AS records are correctly attributed to the right deductor.

Per-Row Financial Year Detection

Unlike most upload systems that assign a single FY to an entire file, each income record is independently assessed — its document date determines the correct financial year (April to March cycle). This allows a single uploaded file to contain records spanning multiple financial years, each automatically routed to the correct FY bucket without manual intervention.

Delete-Insert Upsert Strategy

Income records use a delete-then-insert upsert strategy keyed on doc_number + cust_id (without FY). This ensures that if the same invoice is re-uploaded with a corrected date (and therefore a different FY), the old record is automatically removed and replaced with the corrected one — no manual SQL cleanup required.

Automated Reconciliation Engine

The reconciliation engine is a pure SQL computation joining income_records and form26as_records by customer TAN. It computes match score as $\min(\text{system_tds}, \text{26as_credit}) / \max(\text{system_tds}, \text{26as_credit}) \times 100$, classifying records as Reconciled ($\geq 99\%$), Likely Match ($\geq 85\%$), Suggested Match ($\geq 50\%$), or Open. All statuses can be manually overridden and stored persistently in the recon_overrides table.

Dynamic TDS Deductible Computation Engine

A core technical feature of the system is the update_tds_deductible_from_section() function — a dynamic, fully automated TDS computation engine that runs after every Books of Accounts upload and every Form 26AS upload. Rather than storing static or hardcoded TDS rates against each customer, the engine performs a real-time lookup against the tds_codes_master table — a global reference table built per Finance Act 2026 covering 69 TDS sections across resident and non-resident payee types. For each customer in the financial year, it identifies the applicable section from the uploaded Form 26AS, retrieves the corresponding base rate, surcharge percentage, and cess percentage from the master, applies the legally correct multiplicative compounding formula ($\text{effective_rate} = \text{base_rate} \times (1 + \text{surcharge}/100) \times (1 + \text{cess}/100)$), evaluates the annual aggregate taxable amount against the section's threshold limit, and computes tds_deductible per invoice only when the threshold is breached. This design ensures that any future rate revision requires only a single update to tds_codes_master — no application code changes are needed — making the system inherently future-proof and compliant with successive Finance Acts.

Entry Source Auto-Detection

The entry_source field is auto-detected from file content — TXT files are classified as 'Sales Register' (Tally/TRACES export format), XLSX files with sales register column indicators (voucher_no, ledger, narration, debit/credit columns) are classified as 'Sales Register', and standard manually-filled XLSX files are classified as 'Manual Entry'. No hardcoded values.

Pagination and Performance

All four heavy query endpoints (Books Summary, Income Records, Reconciliation Records, TDS Summary) implement server-side pagination with configurable page size (default 100, max 500). Eleven MySQL indexes cover the primary join columns (cust_id, fin_year, tan_number, doc_number, file_hash) ensuring query performance at scale.

Dual-Database Architecture (MySQL)

The system employs the database strategy — MySQL serves as the primary transactional database handling all inserts, upserts from the Flask application.

The architecture ensures that the high-frequency transactional operations on MySQL never impact the performance of analytical queries run by Power BI.

8. Use Cases

Use Case 1 — Annual TDS Reconciliation

At the end of each financial year, the tax team uploads the Books of Accounts sales register and Form 26AS PDFs for all customers. The system automatically reconciles all records and highlights any variances, significantly reducing the time required from weeks to hours.

Use Case 2 — Mid-Year TDS Credit Verification

Finance teams can upload quarterly Form 26AS data as it becomes available from TRACES and immediately verify whether the TDS credits appearing in government records match the company's books — enabling early identification of discrepancies before year-end.

Use Case 3 — Customer-Level TDS Dispute Resolution

When a vendor or customer disputes a TDS deduction, the Reconciliation screen allows the finance team to drill into individual customer records, view the System TDS vs 26AS Credit breakdown, and use the Change action to mark disputed records as 'Reserved' or 'Not Liable' with explanatory remarks.

Use Case 4 — Management Reporting

The Power BI Dashboard provides the management team with real-time reconciliation status, % Credit achievement, and variance analysis without requiring access to the operational system — refreshing live from the MySQL database via the ODBC connection.

Use Case 5 — Multi-Year Historical Analysis

The system stores records across financial years independently. The Books of Accounts, TDS Summary, and Reconciliation screens all support FY filtering, enabling year-over-year comparison of TDS deductible vs 26AS credit trends across the customer base.

Use Case 6 — New Customer Onboarding

When a new customer relationship is established, the Customer Master module allows immediate onboarding via Excel upload — PAN, TAN, GSTIN, and status are registered centrally, and the new customer is immediately available for reconciliation against any uploaded Form 26AS data.

9. Conclusion

The Form 26AS TDS Reconciliation System successfully addresses the core challenges of manual TDS reconciliation at Sundaram Finance Limited. By automating the parsing, storage, reconciliation, and reporting of TDS data, the system eliminates the risk of human error inherent in spreadsheet-based workflows and provides a single source of truth for the finance and tax teams.

Over the course of the internship, the system grew from a basic CRUD application into a production-grade tool with intelligent PDF parsing, per-row financial year detection, automated reconciliation classification, multi-format file support, complete audit logging, and a connected Power BI dashboard — all built iteratively based on real testing feedback and stakeholder reviews.

Key achievements of the project include:

- Full automation of the Form 26AS reconciliation lifecycle — from data ingestion to status classification
- Support for multiple financial years, multiple customers, and multiple file formats in a single interface
- Zero-hardcoding design — financial year, entry source, and reconciliation status are all dynamically computed from actual data
- Production-quality data integrity features — SHA-256 deduplication, upsert logic, per-row FY validation, and unreferenced entry detection
- A Power BI executive dashboard with three pages, drill-through navigation, and conditional formatting providing finance-grade visual analytics

The system is fully functional and has been demonstrated to and approved by the data platform team at Sundaram Finance Limited. Future enhancements could include user authentication with role-based access control, background job processing for large file uploads, automated email alerts for high-variance customers, and integration with the company's core banking system for real-time data sync.

— *End of Report* —

Author	Sanjana Ravishankar
Organization	Sundaram Finance Limited
Report Date	July 2026
Version	1.0 — Final